

Supplementary Material

Identifying Structural Vulnerabilities in Healthcare Performance Measurement: A Decision Support Framework for Detecting Gaming Behavior

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This document provides supplementary variable lists referenced in the main manuscript. Complete analysis code and data are available at <https://github.com/dustoff06/VA>.

Additional File 1: Complete SAIL Variable List

Table 1: Complete list of SAIL (Strategic Analytics for Improvement and Learning) performance measures used in the analysis.

Measure Name
All Employee Survey (AES) Data Use Engagement Driver
All Employee Survey (AES) Data Use Percentage
All Employee Survey (AES) Sharing Percentage
All Employee Survey (AES) Use Percentage
All Employee Survey (AES) Use Expectations Percentage
Acute Myocardial Infarction (AMI) Risk-Standardized Mortality Rate (RSMR)
Acute Myocardial Infarction (AMI) Risk-Standardized Readmission Rate (RSRR)
Congestive Heart Failure (CHF) Risk-Standardized Mortality Rate (RSMR)
Congestive Heart Failure (CHF) Risk-Standardized Readmission Rate (RSRR)
Chronic Obstructive Pulmonary Disease (COPD) Risk-Standardized Mortality Rate (RSMR)
Chronic Obstructive Pulmonary Disease (COPD) Risk-Standardized Readmission Rate (RSRR)
Pneumonia Risk-Standardized Mortality Rate (RSMR)
Pneumonia Risk-Standardized Readmission Rate (RSRR)
Stroke Risk-Standardized Mortality Rate (RSMR)
Stroke Risk-Standardized Readmission Rate (RSRR)
Acute Care Standardized Mortality Ratio (SMR)
Acute Care 30-Day Standardized Mortality Ratio (SMR30)
Hospital-Wide 30-Day Readmission Rate (HWR)
Hospital-Wide Readmission Rate (Cardiorespiratory Cohort)
Hospital-Wide Readmission Rate (Cardiovascular Cohort)

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Measure Name
Hospital-Wide Readmission Rate (Medicine Cohort)
Hospital-Wide Readmission Rate (Neurology Cohort)
Hospital-Wide Readmission Rate (Surgical Cohort)
Observed Minus Expected Length of Stay (OMELOS)
Adjusted Length of Stay
Hospital Flow Composite
Emergency Department Throughput Composite
Emergency Department Length of Stay (Arrival to Departure, Admitted)
Emergency Department Length of Stay (Arrival to Departure, Discharged)
Emergency Department Length of Stay (Admit Decision to Departure)
Emergency Department Arrival to Admit Decision Time
Percent Left Without Being Seen (LWBS)
Central Line-Associated Bloodstream Infection (CLABSI)
Catheter-Associated Urinary Tract Infection (CAUTI)
Methicillin-Resistant Staphylococcus Aureus (MRSA) Infection
Clostridioides difficile (C. difficile) Infection
Ventilator-Associated Events (IVAC Plus)
Ventilator-Associated Pneumonia
Patient Safety Indicator Composite (PSI-90)
Patient Safety Indicator Average Standardized Score
Postoperative Sepsis Rate
Postoperative Respiratory Failure Rate
Postoperative Pulmonary Embolism or Deep Vein Thrombosis
In-Hospital Complications
Healthcare Effectiveness Data and Information Set (HEDIS-like) Measures
Outpatient Performance Measures (HED90)
Outpatient Performance Measures (HED90 Enhanced)
Outpatient Performance Measures (HED90 Variant)
Electronic Quality Measures (eQM) Cardiovascular Risk Management
Electronic Quality Measures (eQM) Diabetes Composite
Electronic Quality Measures (eQM) Ischemic Heart Disease Composite
Electronic Quality Measures (eQM) Influenza Immunization Composite
External Peer Review Program (EPRP) Global Measures Composite
External Peer Review Program (EPRP) Prevention Composite
External Peer Review Program (EPRP) Behavioral Health Screening
Sepsis Early Management Bundle (SEP-1)
Ambulatory Care Sensitive Condition (ACSC) Hospitalizations
Ambulatory Care Sensitive Condition (ACSC) Hospitalizations (Observed/Expected Ratio)
Primary Care Wait Time
Specialty Care Wait Time
Mental Health Wait Time
Days Waited for Urgent Appointment (Primary Care Medical Home, PCMH)
Days Waited for Urgent Appointment (Primary Care)
Primary Care Medical Home (PCMH) Care Coordination
Specialty Care Coordination
Primary Care Medical Home (PCMH) Stress Discussed
Primary Care Medical Home (PCMH) Timely Appointment, Care, and Information

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Measure Name
Specialty Care Timely Appointment, Care, and Information
Continuity of Care
Comprehensiveness of Care
Care Transition (Inpatient)
Care Transition (HCAHPS)
Hospital Consumer Assessment of Healthcare Providers and Systems (HCAHPS) Overall Rating
Hospital Consumer Assessment of Healthcare Providers and Systems (HCAHPS) Care Transition
Patient Satisfaction
Experience of Care
Overall Rating of Hospital (Inpatient)
Overall Rating of Primary Care Providers
Overall Rating of Specialty Care Providers
Population Coverage
Physician Capacity
Advanced Practice Provider (APP) Capacity
Registered Nurse Turnover Rate
Efficiency (Inverse Stochastic Frontier Analysis, 1/SFA)
Call Center Speed to Answer (seconds)
Call Center Abandonment Rate
New Primary Care Appointments Completed Within 30 Days
New Specialty Care Appointments Completed Within 30 Days
New Mental Health Appointments Completed Within 30 Days
Urgent Care Appointment Availability (Primary Care)
Routine Care Appointment Availability (Primary Care)
Urgent Care Appointment Availability (Specialty Care)
Routine Care Appointment Availability (Specialty Care)
Primary Care New Patient Wait Time (≤ 30 Days)
Specialty Care New Patient Wait Time (≤ 30 Days)
Mental Health New Patient Wait Time (≤ 30 Days)
Admission Reviews Criteria Met
Continued Stay Reviews Criteria Met
Organizational Health Index (OHI) Current Score
Organizational Health Index (OHI) Change Score
Overall Job Satisfaction
Satisfaction with Organization
Recommend Organization as a Good Place to Work

Additional File 2: Complete VAC3 Variable List

Table 2: Complete list of VAC3 (Veterans Affairs Centralized Clinical and Administrative Clarity) variables used in the analysis.

Measure Name
Average Time Spent in Emergency Department Before Discharge
Average Time Spent in Emergency Department Before Being Seen by a Healthcare Professional
Average Time Spent in Emergency Department From Admit Decision to Inpatient Transfer
Average Time Spent in Emergency Department Before Inpatient Admission
Median Time in Emergency Department Before Discharge
Median Time to Admit to Hospital
Percent of Patients Leaving Emergency Department Without Being Seen
Catheter-Associated Urinary Tract Infection (CAUTI)
Central Line-Associated Bloodstream Infection (CLABSI)
Methicillin-Resistant Staphylococcus Aureus (MRSA) Infection
Clostridioides difficile Infection
Hospital-Acquired Infection Composite (Standardized Infection Ratio Measures)
Pressure Ulcer Rate
Postoperative Respiratory Failure Rate
Postoperative Pulmonary Embolism or Deep Vein Thrombosis
Postoperative Sepsis Rate
Postoperative Wound Dehiscence
Accidental Puncture or Laceration During Medical Treatment
Collapsed Lung Due to Medical Treatment
Central Venous Catheter-Related Bloodstream Infection Rate
Postoperative Hip Fracture Rate
Bleeding or Bruising During Surgery
Deaths Among Patients With Treatable Serious Complications
Serious Complications Composite (PSI-90)
30-Day Mortality Rate for Acute Myocardial Infarction (AMI)
30-Day Mortality Rate for Chronic Obstructive Pulmonary Disease (COPD)
30-Day Mortality Rate for Heart Failure (HF)
30-Day Mortality Rate for Pneumonia (PN)
30-Day Readmission Rate for Acute Myocardial Infarction (AMI)
30-Day Readmission Rate for Heart Failure (HF)
30-Day Readmission Rate for Chronic Obstructive Pulmonary Disease (COPD)
30-Day Readmission Rate (Hospital-Wide)
30-Day Readmission Rate for Hip and Knee Replacement
30-Day Excess Days in Acute Care for AMI
30-Day Excess Days in Acute Care for HF
30-Day Excess Days in Acute Care for Pneumonia
Patients Reporting Doctors Always Communicated Well
Patients Reporting Nurses Always Communicated Well
Patients Reporting Staff Always Explained Medications
Patients Reporting Hospital Always Clean
Patients Reporting Hospital Always Quiet

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Measure Name

Patients Reporting Pain Always Well Controlled
Patients Reporting They Always Received Help When Needed
Patients Reporting They Received Discharge Information
Patients Strongly Agree They Understood Care at Discharge
Patients Rating Hospital 9 or 10 (Overall Rating)
Patients Definitely Recommend the Hospital
Hospital Cleanliness and Quiet Composite
Hospital Rating and Recommendation Composite
Care Coordination
Provider Communication Quality
Overall Provider Rating
Clerks and Reception Staff Courtesy and Respect
Ease of Understanding Provider Explanations
Timely Response to Medical Questions
Ability to Get Same-Day Answers to Questions
Ability to Get Care Quickly When Needed
Access to Care During Evenings, Weekends, or Holidays
Provider Knowledge of Patient Medical History
Provider Listening Carefully to Patients
Provider Showing Respect to Patients
Provider Spending Enough Time With Patients
Discussion of Medications With Patients
Discussion of Mental Health or Substance Use Concerns
Discussion of Stress and Personal Concerns
Discussion of Health Goals
Influenza Vaccination Rate (Patients)
Healthcare Worker Influenza Vaccination Rate
COVID-19 Vaccination Rate (Healthcare Personnel)
Breast Cancer Screening
Colorectal Cancer Screening (Ages 45–75)
Cervical Cancer Screening
Chlamydia Screening
Diabetes Blood Pressure Control (< 140/90)
Diabetes Hemoglobin A1c Poor Control (> 9% or Not Done)
Annual Hemoglobin A1c Testing
Diabetes Nephropathy Screening
Retinal Exam for Diabetic Patients
Kidney Health Evaluation for Patients With Diabetes
Statin Therapy for Cardiovascular Disease
Statin Adherence for Cardiovascular Disease
Alcohol Use Screening
Alcohol Brief Intervention Offered
Alcohol Brief Intervention Provided
Tobacco Use Screening
Tobacco Use Treatment Provided During Hospital Stay
Tobacco Cessation Counseling Provided
Tobacco Cessation Medication Offered
Tobacco Cessation Referral Offered

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Measure Name
Early Management Bundle for Severe Sepsis or Septic Shock
Appropriate Follow-Up Interval for Normal Colonoscopy
Primary Care Medical Home (PCMH) Timely Care and Information
Appointment Availability for Routine and Urgent Care
Follow-Up Communication of Test Results
Access to After-Hours Care Information